Logs traces and privacy

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Ask what each field is for and which private values must stay off the emitted path.

Teach privacy-correct preservation, controlled metadata and the difference between declared and implemented controls.

Keep one habit: Model a minimal record → inspect emitted placeholders → distinguish spans from averages → assess fresh mixed fields → specify access and retention controls.

MAIN LESSON · SECTION 1

Prepare the connected service investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of logs traces and privacy

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Failure budgets and service objectives. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of failure budgets and service objectives

Reference example, not the only acceptable wording: Teacher explanation: Service objectives define measured reliability over a stated interval.

A failure budget is meaningful only after success and eligible requests are defined. For a fictional objective at least 98% of 200 eligible requests must pass all contract checks. Allowed failures are 2%*200=4.

Count every eligible failure once; report excluded requests separately.

Without opening the old lesson, explain one key idea from Integration tests and fixtures. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of integration tests and fixtures

Reference example, not the only acceptable wording: Teacher explanation: Integration tests exercise connected components and failure propagation. Fixtures supply controlled **input** records and mocked service outcomes.

Trace request validation, session permission, lookup and **output** validation together. Include absent sources and failing dependencies; verify that denied requests never invoke restricted lookup.

Without opening the old lesson, explain one key idea from Least privilege and tool scopes. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of least privilege and tool scopes

Reference example, not the only acceptable wording: Teacher explanation: Least privilege grants only capabilities required by the task. Tool access should include resource and operation scope, not merely a broad tool name.

A reporting assistant may lookup public rooms but may not write records, send messages or read teacher notes. Check actor, operation and resource before executing a proposed call.

Pictured end goal

Explain logs traces and privacy using the deciding step in this fictional service.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Teach the learning cycle

Retrieval answer guidance

Failure budgets and service objectives

Integration tests and fixtures

Least privilege and tool scopes

Use the three-session teaching cycle above, with extra time as needed. Logs support diagnosis, but excessive recording can expose private content. **Trace ID**s link stages without storing complete sensitive **input**s. Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions. Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of logs traces and privacy

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.
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MAIN LESSON · SECTION 2

Teach diagnostic usefulness, minimisation and unimplemented controls together

> Make the privacy-correct preservation rule explicit throughout the lesson.

Definitions and complete allowed-field policy

Logs support diagnosis, but excessive recording can expose private content. **Trace ID**s link stages without storing complete sensitive **input**s.

For this policy log **trace ID**, stage, status, duration and **error category**; do not log passcodes, student text or teacher answers. Restrict access and declare retention.

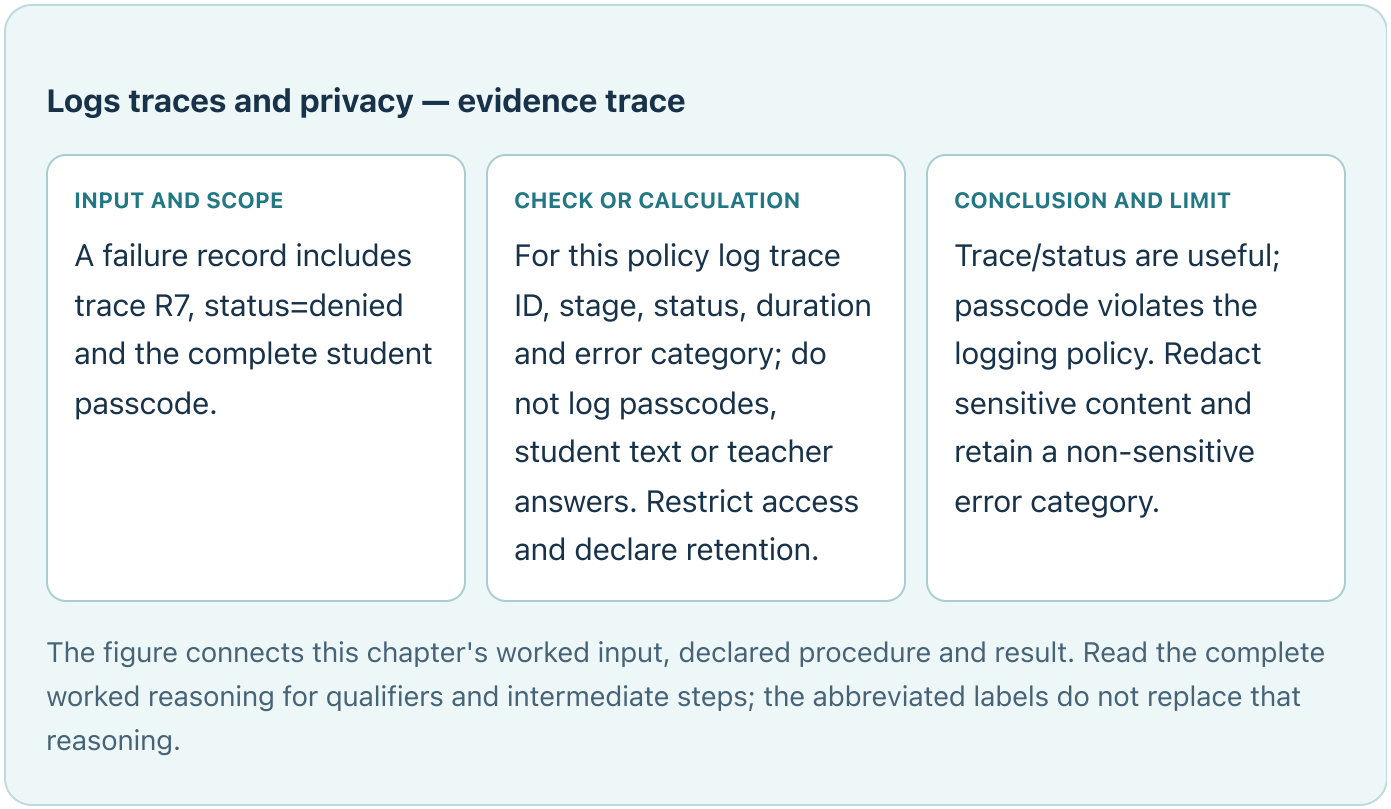
Main classroom logging policy	Purpose or constraint
trace_id	Connect events for one request; application-issued identifier, still potentially linkable
stage	Controlled category: request, authorize, lookup or validate
status	Controlled category: ok, denied or review
duration_ms	Supplied nonnegative integer duration; declare the span being described
error_category	Controlled non-sensitive category, not an arbitrary copied exception message
Excluded fields	Passcodes, raw student question/essay, teacher-answer text and names
Pictured retrieval extension	Its explicitly approved source_id metadata, such as C8, belongs only on lookup events
Access and retention	Declared classroom reviewer and deletion after the exercise; implementation not supplied
Scope	Fictional placeholders and mock records only; no actual private data or log store

Main field decisions and corrected preservation reasoning

Main R7 fixture field	Decision under the declared policy	Reason
trace_id R7	Keep controlled identifier	Links stages without the full request
stage authorize; status denied	Keep controlled categories	Explains the stopping point
duration 3 ms, error permission_denied	Keep supplied mock metadata	Duration/category have declared types and purpose
Complete passcode value	Exclude field and value	Not needed for this diagnostic record
Student text or teacher answer	Exclude field and value	Outside the stated logging policy
Later comparison artifact	Keep sanitized diagnostic record	Honest history does not require retaining private raw content

Reject the raw passcode field; retain allowed non-sensitive diagnostic categories. Do not demonstrate removal by printing the raw mock first. Require value validation as well as field allowlisting, and note that R17/C8 can be linked through other records. The 240 ms value is supplied for one request span, not a measured service average; nested spans are not automatically additive.

Source IDs are permitted only by the pictured retrieval policy, not an unannounced global expansion. The code declares access/retention but does not enforce them. Preserve sanitized failures without retaining real student questions or teacher answers.



Guided: Trace/status are useful; passcode violates the logging policy. Redact sensitive content and retain a non-sensitive **error category**.

Qualification: A **trace ID** can still be linkable; **minimisation** and role-controlled access remain necessary.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

A failure record includes trace R7, status=denied and the complete student passcode.

Trace/status are useful; passcode violates the logging policy. Redact sensitive content and retain a non-sensitive **error category**.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s. The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain a policy-compliant diagnostic record without passcodes, student text or teacher answers, so a later revision can be compared honestly.

Value validation, linkability and control-limit probes

Logging probe	Expected outcome or limit
Main R7 metadata only	Keep declared trace/stage/status/duration/error fields
Raw passcode, question or teacher text added	Violates policy; omit from emitted diagnostic record
Arbitrary private message in error_category	Reject; use declared non-sensitive categories
Boolean or negative duration	Reject under exact nonnegative-integer rule
C8 source ID attached to lookup event	Permitted only under the explicit pictured retrieval extension
Source ID attached to unrelated event	Reject in the local builder’s contract
A record has only allowed field names	Still validate values and purpose; identifiers can be linkable
Access/retention policy is printed	Declaration only; does not prove access enforcement or deletion occurred
One240 ms request fixture	Not the service mean, percentile or future latency guarantee

Separate pictured trace/source/span scope

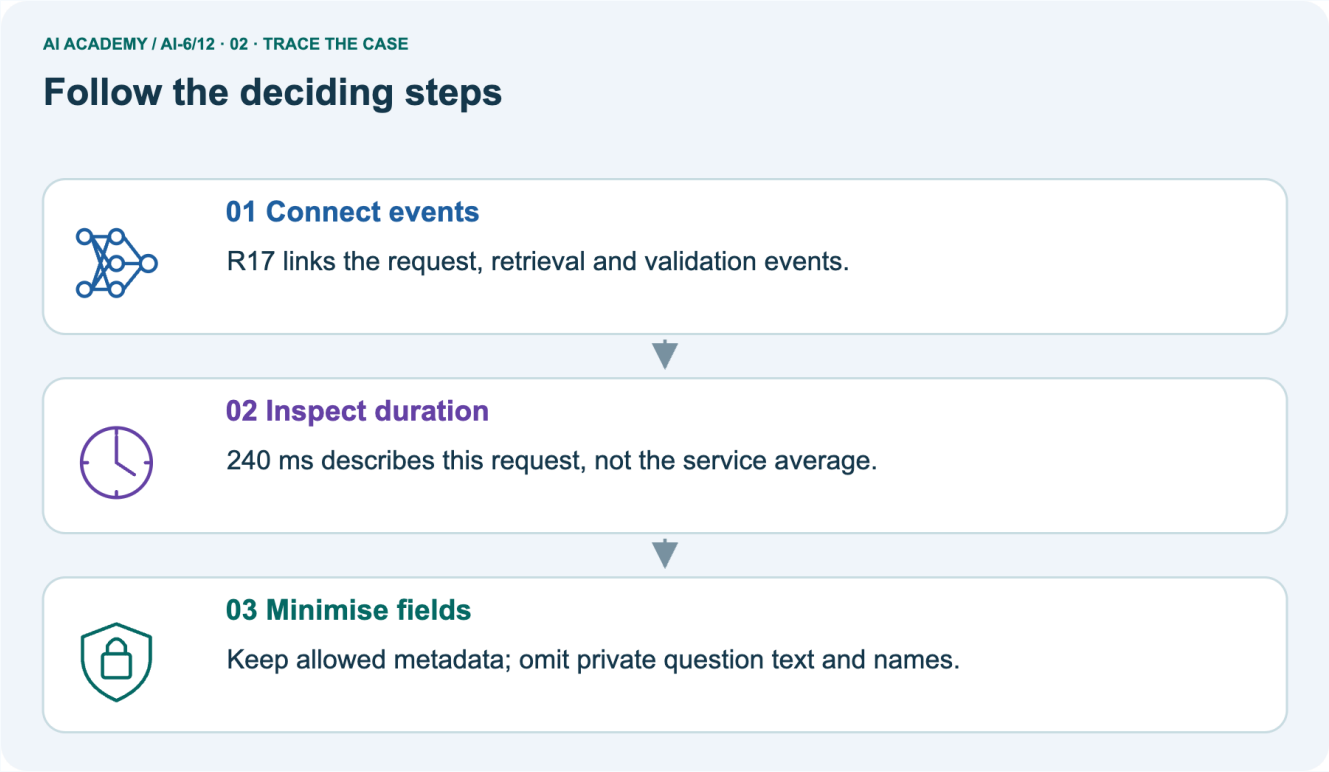
Trace without storing private text. A learner request is retrieved and checked under **trace id**entifier R17.

R17 is the request **trace ID**; C8 is the retrieved source ID. Allowed fields also include validation pass and duration 240 ms. Raw question text is excluded.

The main denied request uses trace R7. The picture is a separate request R17 with retrieved source C8 and supplied request duration 240 ms. R17 connects related events; C8 names the source under this pictured metadata policy; 240 ms describes one request span, not the service average. The local extension uses mock lookup 80 ms and validation 20 ms spans linked to R17.

They can be nested inside the request span, so do not add them to 240 as independent extra work. No timing is actually measured by the example.

Picture trace and full-question violation key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Connect events	R17 links the request, retrieval and validation events.
2. Inspect duration	240 ms describes this request, not the service average.
3. Minimise fields	Keep allowed metadata; omit private question text and names.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Logs record events; traces connect related events. Useful debugging fields should remain within the declared privacy policy.

Trace identifiers can be linkable too; specify their access, retention and deletion rules.

Change: A developer logs the full learner question.

Prediction: That additional field violates this logging policy.

Reason: Debugging convenience does not authorise storing more personal information.

One trace links events without carrying the learner question

EVENT	PERMITTED METADATA	WHAT REMAINS EXCLUDED
Request span	R17, request, ok, supplied240 ms	Raw question and learner name
Lookup span	R17, lookup, ok, supplied80 ms, source C8 under declared extension	Retrieved teacher-answer text
Validation span	R17, validate, ok, supplied20 ms	Raw answer or arbitrary private error content

Supplied classroom spans, not real timing observations. A shared identifier is useful and linkable; it is not a guarantee of anonymity.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Trace ID	An identifier connecting events within an operation
Minimisation	Keeping only data needed for the declared purpose
Error category	A non-sensitive label for a failure class

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

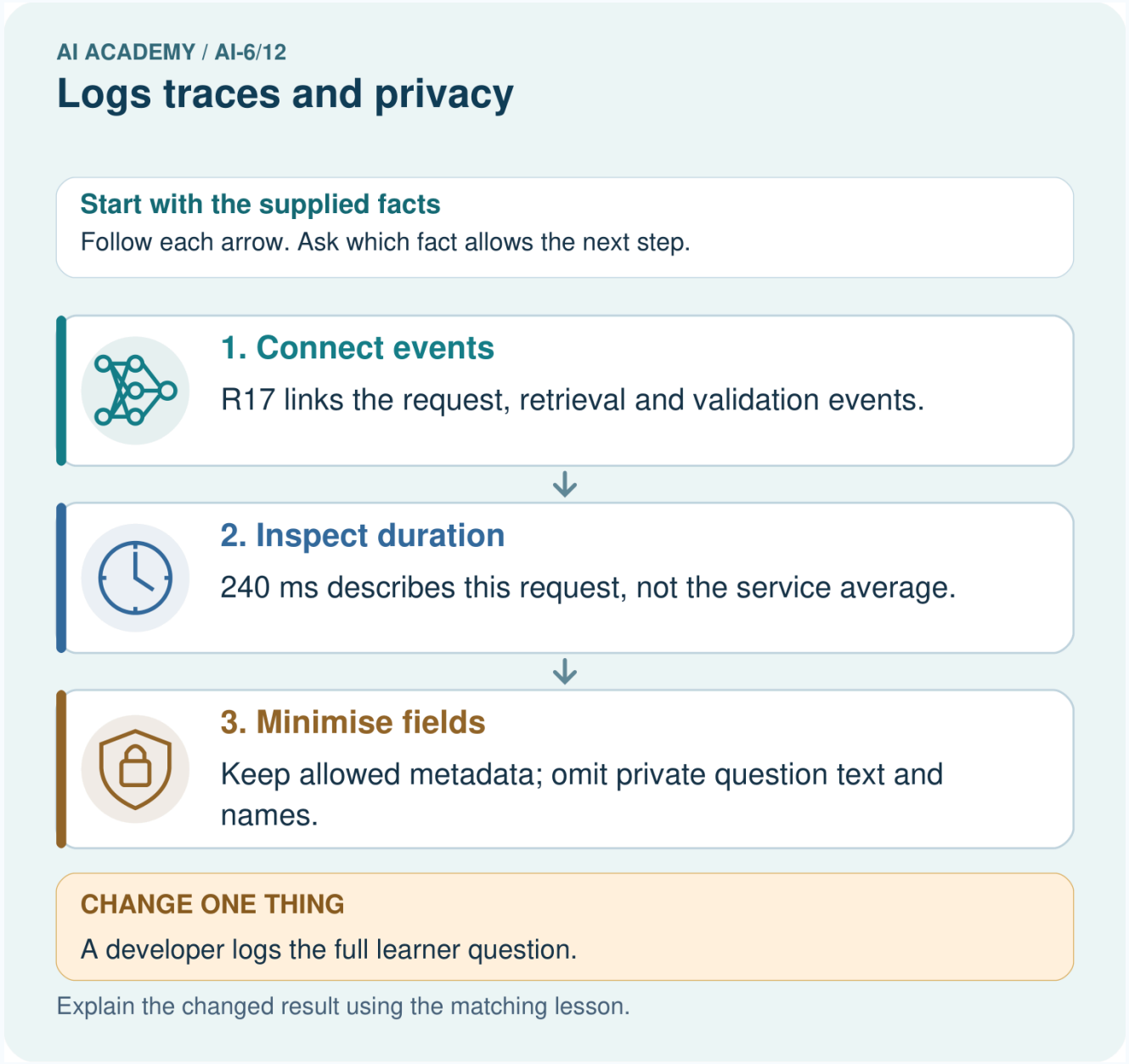
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Logs traces and privacy

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: A failure record includes trace R7, status=denied and the complete student passcode. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Trace/status are useful; passcode violates the logging policy. Redact sensitive content and retain a non-sensitive **error category**.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Logs traces and privacy. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A **trace ID** can still be linkable; **minimisation** and role-controlled access remain necessary.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Identify the purpose of R17, C8 and 240 ms in the trace. Which field is deliberately omitted, and why would logging the whole question violate this design?

R17 links the request, retrieval and validation events. 240 ms describes this request, not the service average. Keep allowed metadata; omit private question text and names.

Changed case: That additional field violates this logging policy. Debugging convenience does not authorise storing more personal information.

R17 links events for one request, C8 identifies the retrieved source, and 240 ms is this request's duration.

Changed-case reasoning: That additional field violates this logging policy. Debugging convenience does not authorise storing more personal information.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main R7: retain allowed metadata and non-sensitive error category; exclude passcode and raw content. Picture R17 links events, C8 source ID allowed only in that retrieval policy, 240 ms is one supplied request span. Independent mixed record: keep available stage lookup, duration 12 ms, traceT8; exclude user essay and teacher guide.

Other required metadata is not supplied: do not invent it or claim a complete validated record. Require access/retention/linkability limits and sanitized evidence preservation.

Diagnose the current misconceptions

The worked result establishes every future case: A **trace ID** can still be linkable; **minimisation** and role-controlled access remain necessary.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.

What the response shows	Next teaching action
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new service scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Logs contain user essay, teacher guide, stage=lookup, duration=12 ms and trace T8. Select allowed fields.

Teacher explanation: Keep stage, duration and trace; exclude essay and teacher content. Authorisation to **process** data does not imply permission to retain it in logs.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: Logs support diagnosis, but excessive recording can expose private content. **Trace ID**s link stages without storing complete sensitive **input**s.

For this policy log **trace ID**, stage, status, duration and **error category**; do not log passcodes, student text or teacher answers. Restrict access and declare retention.

W2 Trace the worked case

A failure record includes trace R7, status=denied and the complete student passcode. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Trace/status are useful; passcode violates the logging policy. Redact sensitive content and retain a non-sensitive **error category**.

W3 Audit the limit

A report omits this qualification: A **trace ID** can still be linkable; **minimisation** and role-controlled access remain necessary. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: A **trace ID** can still be linkable; **minimisation** and role-controlled access remain necessary.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A **trace ID** can still be linkable; **minimisation** and role-controlled access remain necessary.

The new case should test this particular **mechanism** rather than merely restate the original result.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Guided and exit questions

Guided: Trace/status are useful; passcode violates the logging policy. Redact sensitive content and retain a non-sensitive **error category**.

Qualification: A **trace ID** can still be linkable; **minimisation** and role-controlled access remain necessary.
</details>

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Logs contain user essay, teacher guide, stage=lookup, duration=12 ms and trace T8. Select allowed fields.

Teacher explanation: Keep stage, duration and trace; exclude essay and teacher content. Authorisation to **process** data does not imply permission to retain it in logs.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: Keep stage, duration and trace; exclude essay and teacher content. Authorisation to **process** data does not imply permission to retain it in logs. Require intermediate reasoning and the stated scope.

Fresh retry: Log contains token, **trace ID** and timeout category. Expected reasoning: Keep trace/category, redact token. Retention and access rules still apply.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

Connect, support and revisit service design

> Use saved evidence and boundary decisions to choose the next step.

Project and next-week bridge

Add a labelled logs traces and privacy **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of logs traces and privacy.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: State machines for bounded agents. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Delayed return and support record

Suggested return points in this level: ai-6/13, ai-6/15, ai-6/18. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of logs traces and privacy

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: A failure record includes trace R7, status=denied and the complete student passcode. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: Trace/status are useful; passcode violates the logging policy. Redact sensitive content and retain a non-sensitive **error category**.

Check every learner before discussion: Use the worked case for Logs traces and privacy. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A **trace ID** can still be linkable; **minimisation** and role-controlled access remain necessary.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: A developer logs the full learner question. Require a **prediction** and a reason before comparing. Expected result: That additional field violates this logging policy.

Reason: Debugging convenience does not authorise storing more personal information.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

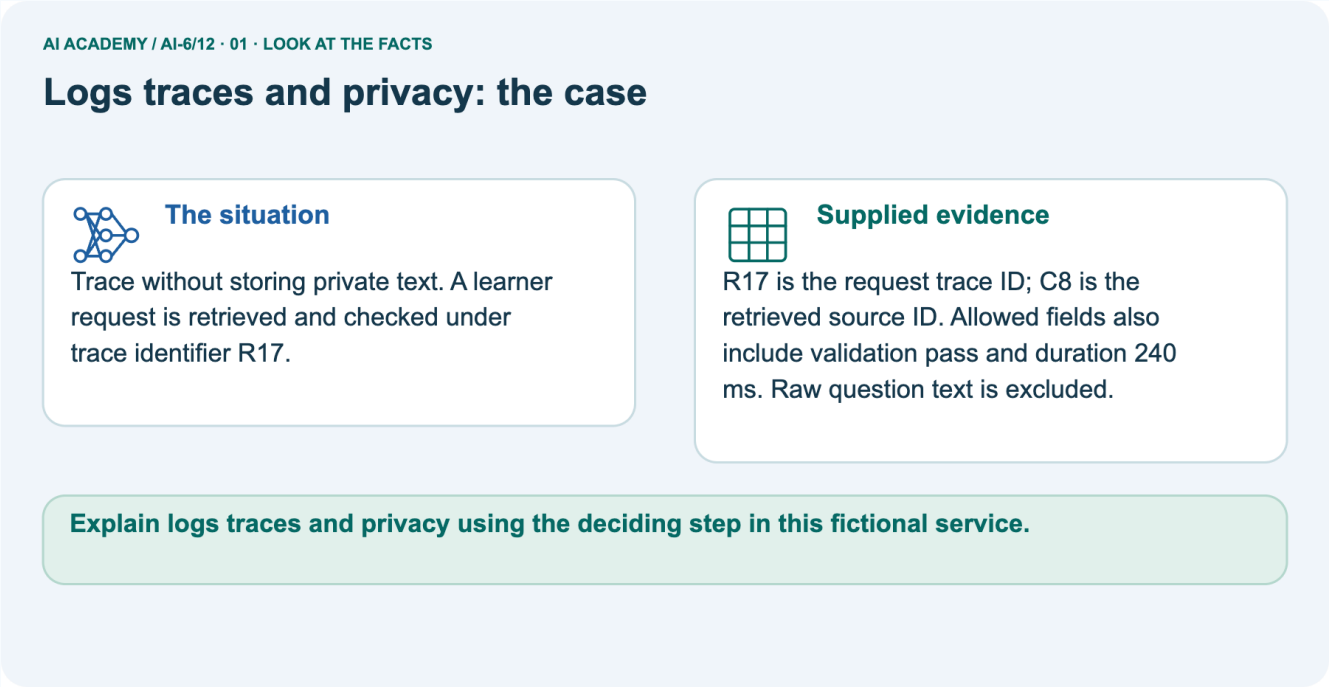
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Logs traces and privacy: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

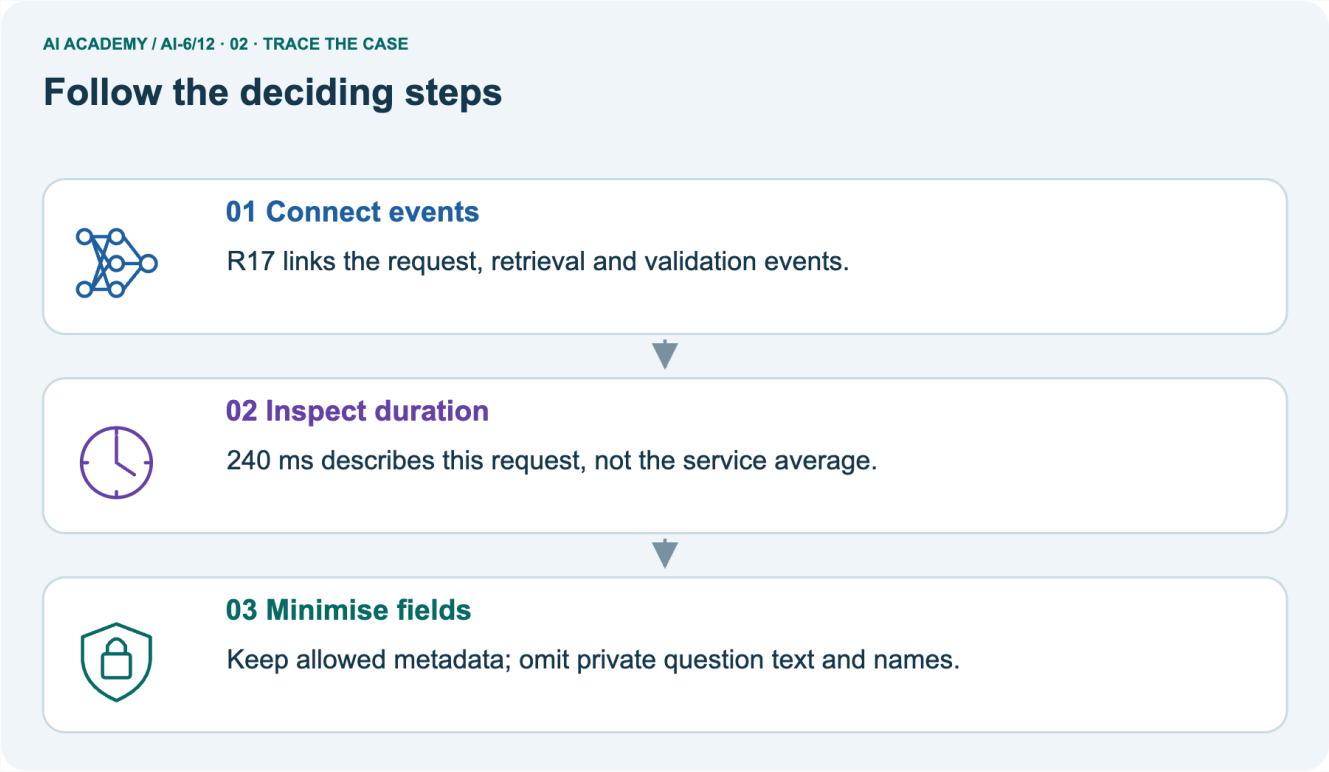
Trace without storing private text. A learner request is retrieved and checked under **trace id**entifier R17. Supplied facts: R17 is the request **trace ID**; C8 is the retrieved source ID.

Allowed fields also include validation pass and duration 240 ms. Raw question text is excluded.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



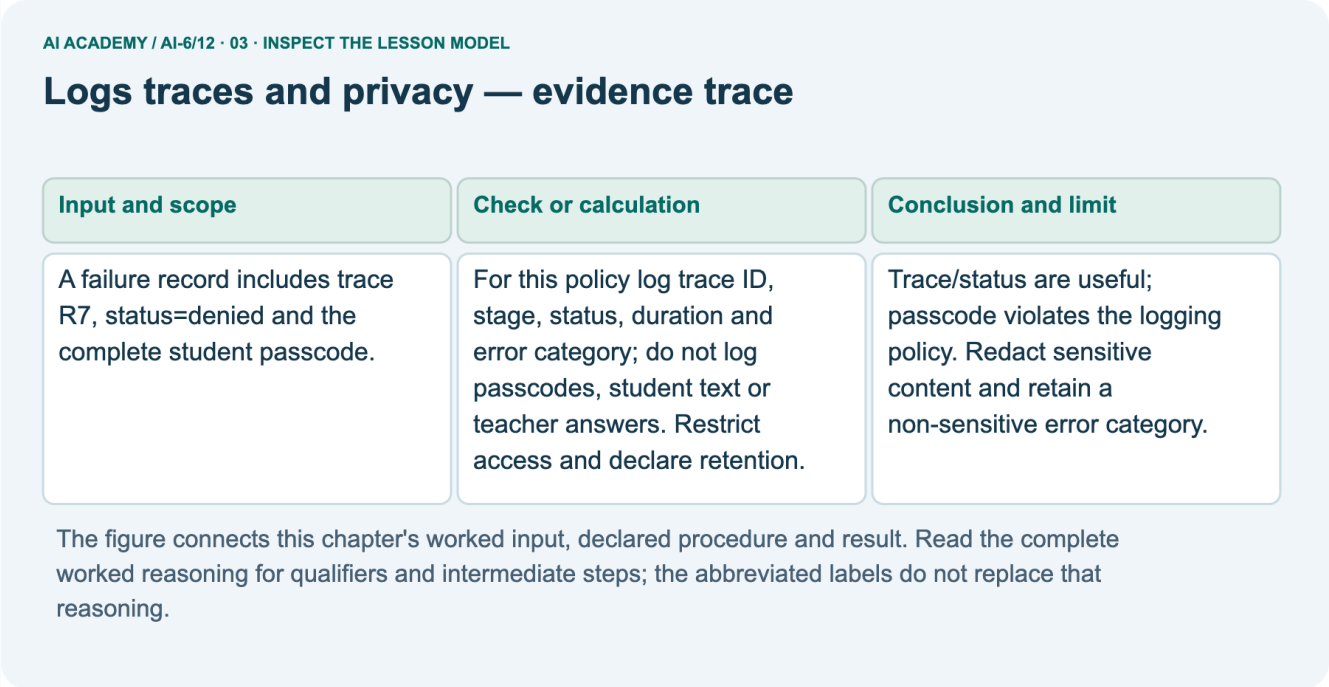
Follow the labelled picture with the matching explanation. Open the vector diagram

Connect events: R17 links the request, retrieval and validation events. → Inspect duration: 240 ms describes this request, not the service average. → Minimise fields: Keep allowed metadata; omit private question text and names.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Logs traces and privacy — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Logs traces and privacy — **evidence** trace.

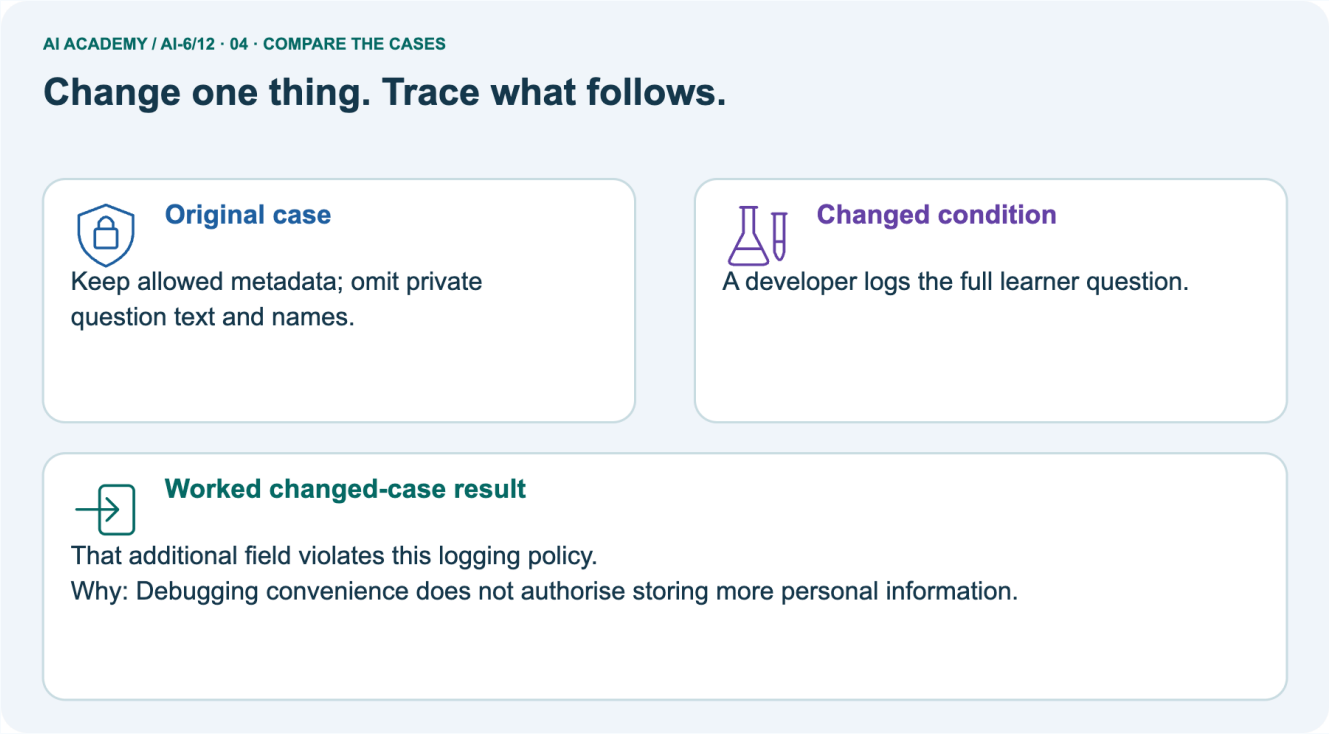
A failure record includes trace R7, status=denied and the complete student passcode. | For this policy log **trace ID**, stage, status, duration and **error category**; do not log passcodes, student text or teacher answers.

Restrict access and declare retention. | Trace/status are useful; passcode violates the logging policy. Redact sensitive content and retain a non-sensitive **error category**.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Keep allowed metadata; omit private question text and names. Changed condition: A developer logs the full learner question. Expected result: That additional field violates this logging policy.

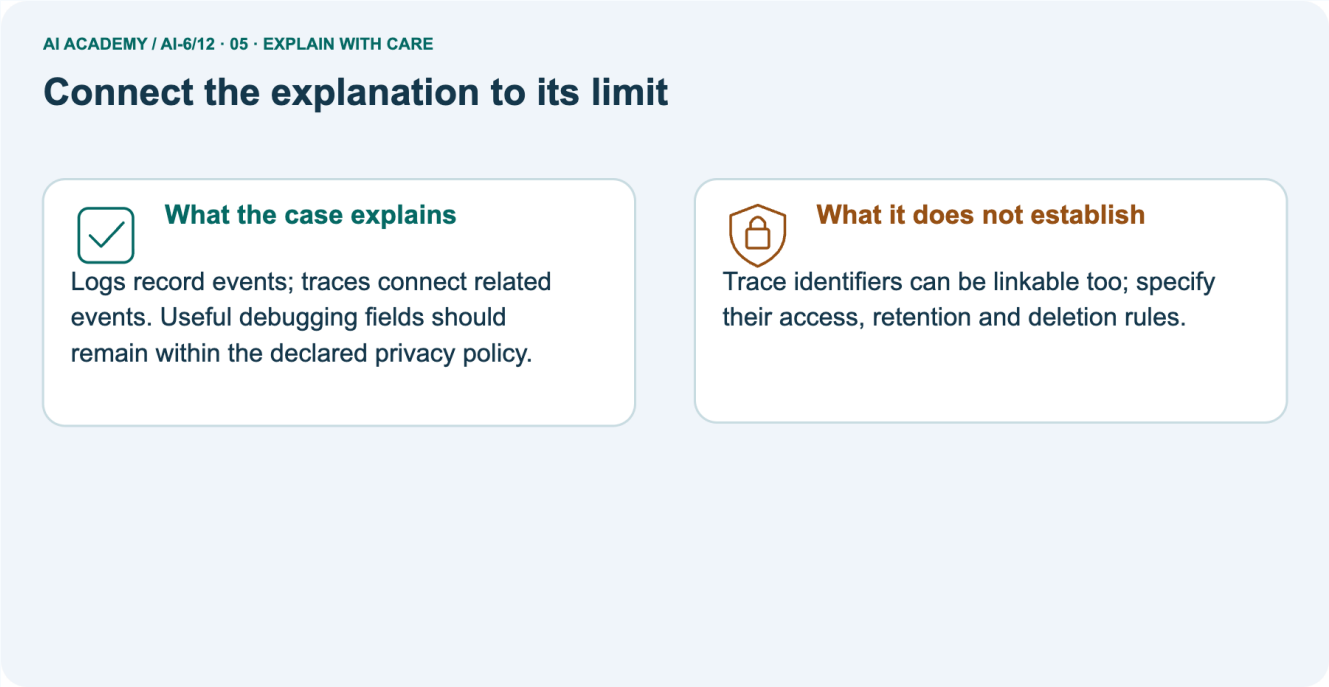
Reason: Debugging convenience does not authorise storing more personal information.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Debugging convenience does not authorise storing more personal information.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

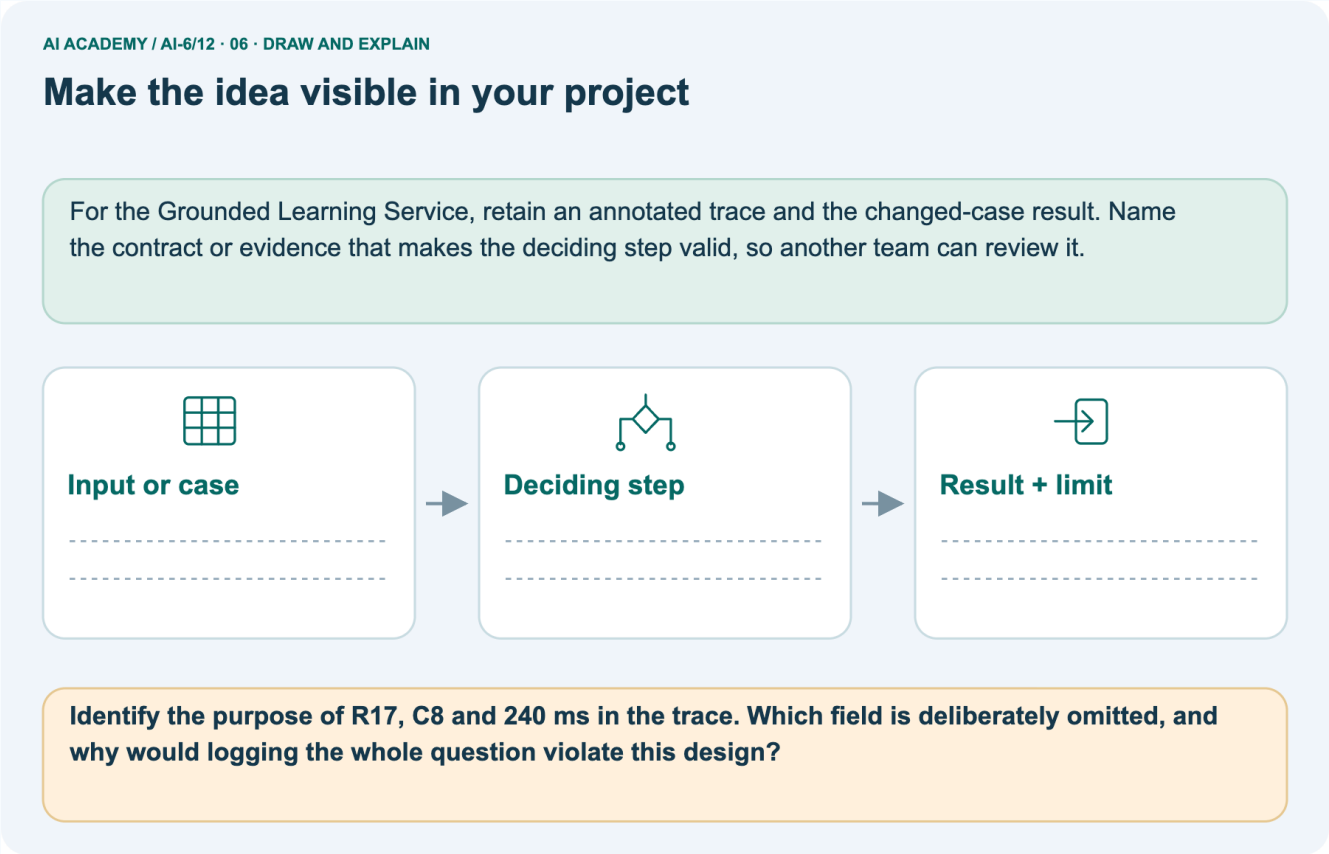
Logs record events; traces connect related events. Useful debugging fields should remain within the declared privacy policy. Limit: **Trace id**entifiers can be linkable too; specify their access, retention and deletion rules.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: **Trace id**entifiers can be linkable too; specify their access, retention and deletion rules.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Identify the purpose of R17, C8 and 240 ms in the trace. Which field is deliberately omitted, and why would logging the whole question violate this design?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: R17 links the request, retrieval and validation events. 240 ms describes this request, not the service average. Keep allowed metadata; omit private question text and names.

Changed case: That additional field violates this logging policy. Debugging convenience does not authorise storing more personal information.

R17 links events for one request, C8 identifies the retrieved source, and 240 ms is this request's duration.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Logs traces and privacy: the case

02. Follow the deciding steps

03. Logs traces and privacy — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.